Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building web interfaces, REST integrations, and reliable workflow platforms. Experienced in React, AngularJS, TypeScript, Java, Python, TCP/IP, JSON, SQL, design patterns, automated testing, requirements traceability, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operator-facing dashboards for 500+ users, translating workflow requirements into secure interfaces and indexed queries that reduced response latency from 1.2 seconds to 350 milliseconds.
- Implemented fault-tolerant background workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while strengthening requirements traceability, observability, and reliable production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React interfaces and Flask REST services for secure upload, routing, status, and history workflows, translating instructor requirements into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated unit, functional, and integration validation, AWS RDS persistence, and error handling, reducing turnaround 60% while maintaining stable system behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying automated tests, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled system evaluations.
- Participated in cross-functional design reviews and validation experiments, documenting technical decisions and tuning a spatial-temporal architecture that improved closed-loop system stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented REST and ETL integration components using reusable design patterns, AWS Glue, and S3, improving response times 30% and workflow efficiency 68% across complex external-system interfaces.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving engineering confidence in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js interfaces across eight authentication, customer, scheduling, permissions, and notification workflows, synchronizing 2 million+ Azure SQL records through maintainable service boundaries and WebSockets.
- Automated Docker builds, unit, integration, and system tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving traceability and operational reliability.

Skills

Languages & UI: TypeScript, JavaScript, React, AngularJS, Java, Python, C++, HTML5, CSS3, responsive interfaces, component design
Architecture & Integration: REST APIs, JSON, XML, TCP/IP, client-server architecture, microservices, WebSockets, object-oriented programming, SOLID design, design patterns, API integration, workflow engines
Data & Cloud: Azure SQL, SQL Server, PostgreSQL, MySQL, MongoDB, AWS RDS, Azure, AWS, Amazon S3, Blob Storage, relational databases, indexing, query optimization
Quality & Delivery: Git, GitHub, code review, Agile, Scrum, Jenkins, CI/CD, Docker, PyTest, Mockito, unit testing, functional testing, system testing, integration testing, requirements traceability, debugging, documentation, observability

Projects

HMIS Live | Live | *TypeScript, React, REST APIs, Appwrite, Sentry*
- Built healthcare workflows for appointments, records, and consent-based data access using responsive React interfaces, REST APIs, secure authentication, automated notifications, and Sentry monitoring to improve reliability and issue traceability.

PiXMod | Live | *Next.js, AngularJS, TypeScript, MongoDB, Stripe*
- Delivered a responsive AngularJS and Next.js SaaS interface with REST integrations, secure subscription workflows, MongoDB persistence, and reusable components, reliably processing 1,000+ user-edited assets across maintainable product workflows.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a cross-platform workflow application supporting 10+ databases, combining Java services, TypeScript interfaces, schema-aware validation, Docker, and Maven to improve reliable data exploration, system integration, and management.